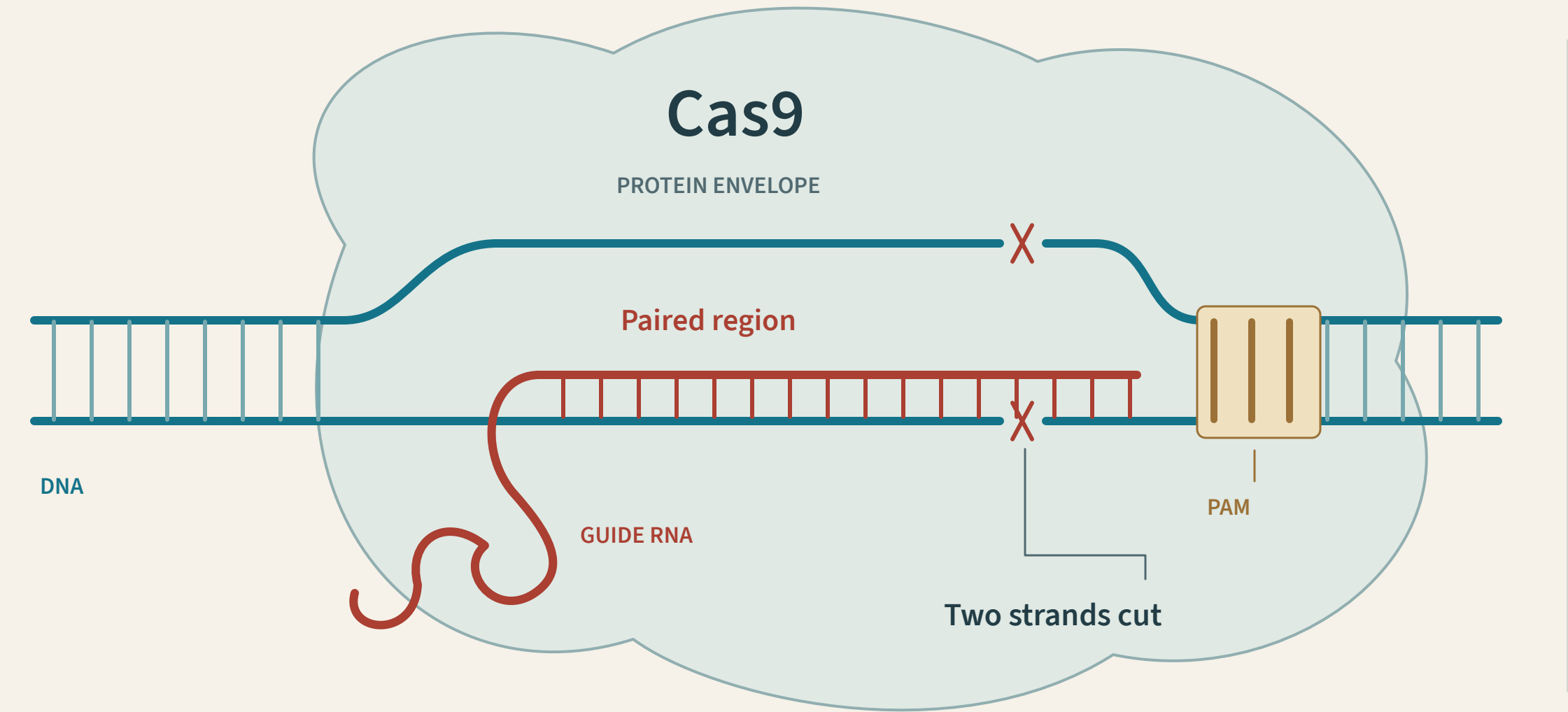


CRISPR–Cas9: targeting is only half the story

RNA-guided recognition locates a break. DNA repair determines what the genome becomes.

01 Recognition is a molecular fit

The guide pairs with one DNA strand while Cas9 holds the locally opened duplex.



A LANDMARK

PAM is adjacent DNA

Cas9 requires a compatible motif next to the target.

A MATCH

The RNA supplies specificity

Complementary pairing positions the nuclease. Similar sites can still be cut.

Strand separation and protein shape are schematic.

02 A cut is the beginning of an edit

The cell repairs the break through competing pathways; these are two simplified outcomes.

